

DAX Functions in Power BI Assignment

You are working as a Data Analyst for a retail company. The management wants interactive Power BI reports using DAX (Data Analysis Expressions) to analyze sales, profit, rankings, and time-based performance. Your task is to create calculated columns, measures, calculated tables, and KPIs using DAX.

Tasks to be performed:

Q1. Create a calculated column Total Price using Quantity and Unit Price.

Answer:

1 Total Price = amazon_store_sales_2022[Quantity]*amazon_store_sales_2022[UnitPrice]										
OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	Total Price	
04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22	
06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8	
08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34	
11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18	
15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4	
18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12	
21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98	
30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4	
31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48	
13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6	
16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33	
24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12	
25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96	
28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	293.79	
06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61	
15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	947.72	
17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	222.36	
19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	2127.04	
22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	841.54	
23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	1162.65	
27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	2409	
04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	318.84	
07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	2993.52	
15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	952.47	
21 April 2022	2009	Beauty	7	399.41	0.13	3300.95	North	Online	2795.87	
30 April 2022	2035	Beauty	6	127.53	0.06	4780.01	North	Online	765.18	
01 May 2022	2035	Beauty	7	281.41	0.02	145.21	Central	Mobile App	1969.87	

Q2. Create a calculated column Discounted Price.

Answer:

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1 Discounted Price = amazon_store_sales_2022[Total Price]-(amazon_store_sales_2022[Total Price]*amazon_store_sales_2022[Discount])
```

OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	Total Price	Discounted Price
04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22	1422.1116
06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8	267.654
08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34	531.8898
11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18	1751.7132
15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4	82.712
18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12	3350.808
21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98	275.2614
30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4	36.852
31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48	1973.3824
13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6	67.07
16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33	495.7839
24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12	2319.84
25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96	1280.664
28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	293.79	235.032
06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61	1612.8497
15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	947.72	777.1304
17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	222.36	195.6768
19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	2127.04	1999.4176
22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	841.54	647.9858
23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	1162.65	941.7465
27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	2409	2360.82
04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	318.84	296.5212
07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	2993.52	2394.816
15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	952.47	819.1242

Q3. Create a measure Average Order Value (AOV).

Answer:

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1 Average Order Value (AOV) = [Total Sales]/[Total Orders]
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1 Total Orders = DISTINCTCOUNT(amazon_store_sales_2022[OrderID])
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1 Total Sales = SUM(amazon_store_sales_2022[SalesAmount])
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Q4. Using DAX, create a measure that calculates cumulative Year-to-Date (YTD) Sales based on the Order Date.

Answer:

```
1 Total YTD Sales = TOTALYTD(SUM(amazon_store_sales_2022[SalesAmount]), amazon_store_sales_2022[OrderDate])
```

Q5. Using DAX, create a measure to rank each product according to its Total Sales, with the highest-selling product ranked first.

Answer:

```
1 Product Rank = RANKX(ALL(amazon_store_sales_2022[ProductID]), SUM(amazon_store_sales_2022[SalesAmount]), , DESC)
```

Q6. Create a calculated column classifying orders as “High Value” if sales > 2000, else “Low Value”.

Answer:

1 Product Value = IF(amazon_store_sales_2022[SalesAmount] > 2000, "High Value", "Low Value")

OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	Total Price	Discounted Price	Product Value
04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22	1422.1116	Low Value
06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8	267.654	High Value
08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34	531.8898	High Value
11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18	1751.7132	High Value
15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4	82.712	High Value
18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12	3350.808	Low Value
21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98	275.2614	High Value
30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4	36.852	High Value
31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48	1973.3824	Low Value
13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6	67.07	High Value
16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33	495.7839	Low Value
24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12	2319.84	High Value
25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96	1280.664	High Value
28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	293.79	235.032	Low Value
06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61	1612.8497	High Value
15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	947.72	777.1304	High Value
17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	222.36	195.6768	High Value
19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	2127.04	1999.4176	High Value
22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	841.54	647.9858	High Value
23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	1162.65	941.7465	High Value
27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	2409	2360.82	Low Value
04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	318.84	296.5212	Low Value
07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	2993.52	2394.816	Low Value
15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	952.47	819.1242	High Value
21 April 2022	2009	Beauty	7	399.41	0.13	3300.95	North	Online	2795.87	2432.4069	High Value
30 April 2022	2035	Beauty	6	127.53	0.06	4780.01	North	Online	765.18	719.2692	High Value
01 May 2022	2035	Beauty	7	281.41	0.02	145.21	Central	Mobile App	1969.87	1476	Low Value

Q7. Create a measure to calculate total sales from the Online channel only.

Answer:

1 Total Sales Online = CALCULATE(SUM(amazon_store_sales_2022[SalesAmount]), amazon_store_sales_2022[SalesChannel] = "Online")

OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	Total Price	Discounted Price
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Q8. Create a measure called "Date Difference" that calculates the number of days between the latest Order Date and today's date using the DATEDIFF function.

Answer:

1 Date Differnece = DATEDIFF(MAX(amazon_store_sales_2022[OrderDate]), TODAY(), DAY)